

SUMMARY

Full-Stack & Backend Engineer (B.Tech CSE '26, CGPA 8.75/10) with hands-on experience architecting production-grade AI applications and distributed microservices using **Node.js, Express.js, TypeScript, Python, MongoDB Atlas, and Docker**. Proven track record designing end-to-end RAG pipelines, low-latency streaming APIs (SSE), and scalable authentication systems. Targeting backend/SWE roles focused on distributed systems, clean API design, and cloud deployments.

EDUCATION

B.Tech in Computer Science Engineering	2022 – 2026
Mohan Babu University, Tirupati, Andhra Pradesh	CGPA: 8.75 / 10.0
Intermediate – MPC	2020 – 2022
Narayana Junior College, Visakhapatnam	88.9%

TECHNICAL SKILLS

Languages	TypeScript, JavaScript (ES6+), Python, Java, SQL, HTML5, CSS3
Backend	Node.js, Express.js, FastAPI, RESTful APIs, Server-Sent Events (SSE), WebSockets
Frontend	React 19, Vite, Tailwind CSS, Framer Motion, Radix UI
Databases	MongoDB Atlas (MQL, Mongoose ODM), Vector Search, Redis, PostgreSQL, SQLite
AI & Systems	RAG (Retrieval-Augmented Generation), Gemini GenAI SDK, Vector Embeddings
DevOps & Cloud	Docker, Render, Vercel, GitHub Actions (CI/CD), AWS (EC2, S3)
Concepts	System Design, Data Structures & Algorithms, OOP Principles, JWT Authentication
Tools	Git, GitHub, Postman, VS Code, Antigravity

PROJECTS

Cortex – Autonomous AI Learning Ecosystem

React 19 · TypeScript · Node.js · Express · MongoDB Atlas · Gemini AI · Docker · Vercel · Render

[Live Demo](#) · [Source](#)

- Architected an adaptive orchestration engine converting unstructured documents into hierarchical academic roadmaps and interactive neural concept maps using **Google Gemini 2.5/Flash**.
- Built an end-to-end **RAG (Retrieval-Augmented Generation) pipeline** leveraging in-process PDF chunking and **3072-dimensional vector embeddings** with MongoDB Vector Search for high-accuracy curriculum grounding.
- Engineered an asynchronous streaming architecture (**Server-Sent Events**) for low-latency interactive AI tutoring, supporting active recall, dynamic code sandboxes (7 languages), and concept visualization.
- Implemented robust multi-tenant authentication featuring role-based access control (**RBAC**) for university admins/students, automated session recovery, and 1-click ephemeral sandbox isolation.
- Containerized microservices using **Docker** deployed on Render, orchestrated SPA delivery via Vercel with automatic rewrites, and enforced strict type safety (**zero TypeScript warnings**).

HireCircle – AI-Powered Blue-Collar Hiring Platform

React 19 · TypeScript · Vite · Tailwind CSS · Gemini GenAI SDK · MediaDevices API

[Source](#)

- Built a mobile-first hiring platform with dual-role authentication (Job Seeker / Employer), supporting phone and email login flows with role-specific theming across **3,100+ lines of TypeScript**.
- Integrated **Google Gemini 3 (Pro & Flash)** via the GenAI SDK for AI-powered video interviews that extract structured profiles (skills, experience, qualifications) using **JSON schema-constrained generation**.
- Engineered a **5-module tab architecture** (Connect feed, Profiles, Applications, Jobs, Settings) with stack-based navigation, full-screen chat interfaces, and AI-generated smart reply suggestions.
- Implemented a social hiring layer (Connect tab) featuring community feeds with voice/photo posts, a **karma-based vouch system**, industry Circles with rate boards, an Academy with video courses, and referral Bounties.
- Designed real-time application tracking with a **6-stage pipeline** (Applied → Viewed → Shortlisted → Interview → Offer → Rejected) and AI-powered candidate-job match scoring with Gemini-generated explanations.

CalcVerse – All-in-One Modular Calculator Platform

Node.js · Express.js · REST API · Docker · Render

[Live Demo](#) · [Source](#)

- Engineered a modular backend using **Node.js/Express** with 15+ REST endpoints, achieving **200ms avg response time** and **35% latency reduction** via request optimization and response caching.

- Designed a **plugin-style computation engine** decoupling each calculation module for independent scaling, unit testing, and maintenance.
- Wrote comprehensive test suites for edge cases and boundary conditions; containerized with **Docker** and deployed on Render with environment-based configuration management.

ACHIEVEMENTS, DSA & IMPACT

- **Data Structures & Algorithms:** Solved **150+ LeetCode problems** across Arrays, Hash Maps, Binary Search, Two Pointers, Stacks, Queues, Sliding Window; applied Big-O time-space complexity optimization.